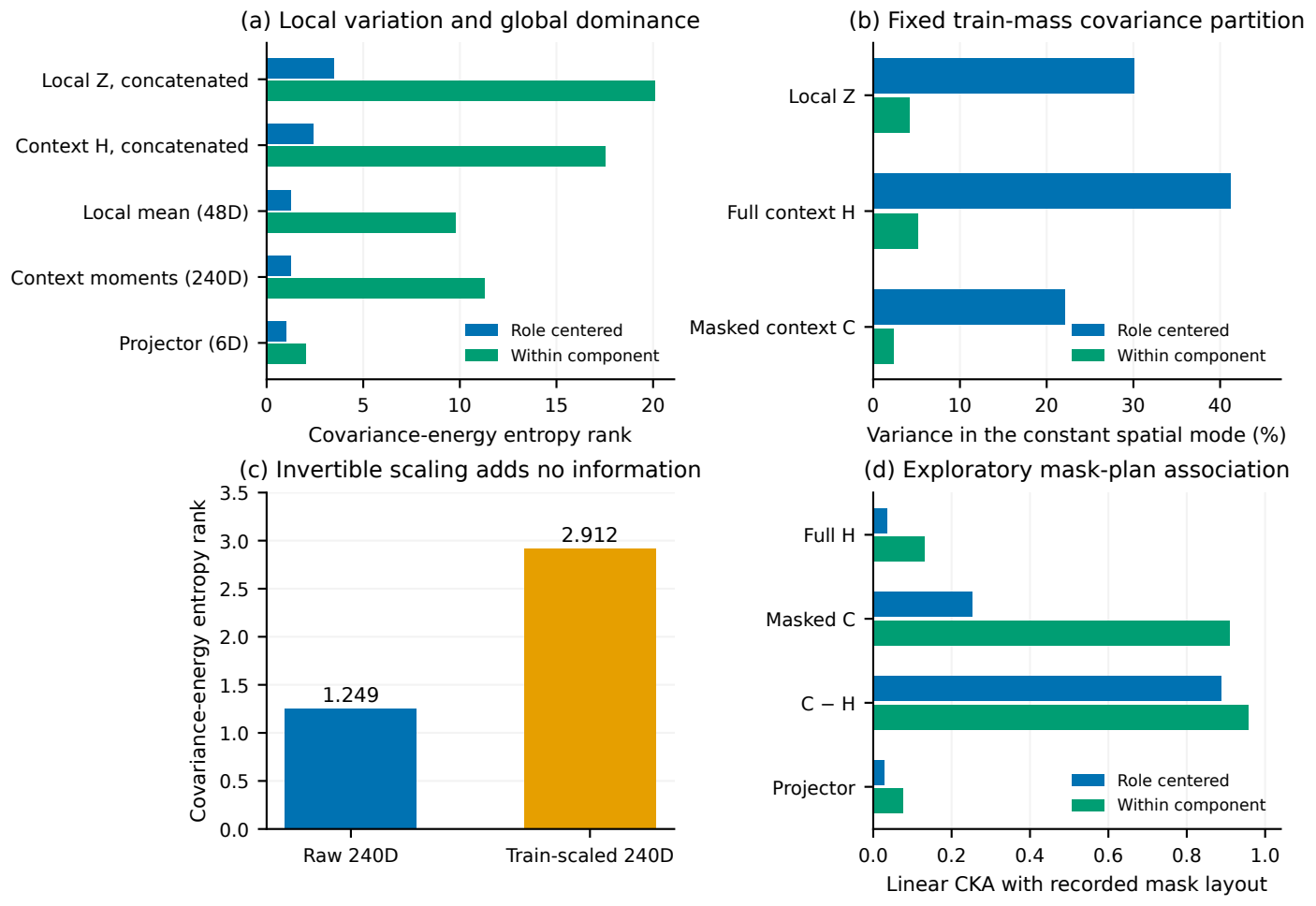
(a) Local variation and global dominance
Local Z, concatenated
Context H, concatenated
Local mean (48D)
Context moments (240D)
Projector (6D)
Role centered
Within component
0
5
10
15
20
Covariance-energy entropy rank
(b) Fixed train-mass covariance partition
Local Z
Full context H
Masked context C
Role centered
Within component
0
10
20
30
40
Variance in the constant spatial mode (%)
(c) Invertible scaling adds no information
Covariance-energy entropy rank
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
1.249
2.912
Raw 240D
Train-scaled 240D
(d) Exploratory mask-plan association
Full H
Masked C
C − H
Projector
Role centered
Within component
0.0
0.2
0.4
0.6
0.8
1.0
Linear CKA with recorded mask layout